

A · SAMPLE CALLING

SHARED INPUT
primer.csv

locus · forward · reverse

GENOME ASSEMBLY

Shared Sassy batch PCR

primer-bounded products → length → calibrated repeats

PAIRED-END ILLUMINA FASTQ

minimap2 → htlib evidence

cached short.mmi · molecule/state reduction

ACCURATE LONG / AMPLICON FASTQ

minimap2 → htlib evidence

cached long.mmi · temporary compressed BAM

direct molecules → shared allele inference

SHARED ALLELE INFERENCE

Common locus calls

locus · repeat count
call status · confidence/evidence
locus sequence, when resolved

*Detection may remain presence-only or unresolved;
a repeat count is reported only when evidence supports it.*

OUTPUT

**MLVA
fingerprint**
sample × loci

REFERENCE-ENABLED ANALYSIS

COMMON CALLS + DATABASE

Multi-locus comparison

Reference comparison
repeat-count distance + sequence evidence

Locus sequence placement
MAFFT query alignment → EPA-ng

Weighted evidence across loci
recovery · discrimination · bootstrap stability

FINAL REFERENCE-ENABLED RESULT

Taxon assignment

assignment status + confidence/evidence
not a single nearest-hit lookup

DATABASE ALSO SUPPORTS CALLING

FASTQ: cached candidates + indexes
minimap2 pipe → htlib → BAM
no text candidate SAM in normal operation

*One global thread budget supports bounded concurrent
samples and deterministic combined output ordering.*

B · REFERENCE DATABASE CONSTRUCTION

INPUTS

taxids.csv
+ primer.csv

NCBI DATASETS

Taxon cohorts
amplicons + QC only

LOCUS RECOVERY

Sassy extraction
multiple-product QC
amplifiability summary

FINALIZE ONCE

candidate bank + .mmi
Deacon index + locus trees
RAxML-NG fixed at 1 thread

REUSABLE REFERENCE DATABASE

**Combined multi-taxon
mlvamaps database**

candidates · indexes · real loci
trees · metadata · provenance